

PLC | SCADA | DCS | DRIVES

The Automation People

PROLIFIC

Training Catalog



Table of Contents

Automation & Process control

SN	Training Module	Level	Days	Participants In a Batch
1	**Programmable Logic Controllers (PLC)** Allen Bradley, Siemens, Modicon, GE Fanuc, Mitsubishi, Messung, Omron	Basic	3	Up to 12
2	**Programmable Logic Controllers (PLC)** Allen Bradley, Siemens, Modicon, GE Fanuc, Mitsubishi, Messung, ABB	Advanced	4	Up to 12
3	**SCADA (HMI)** InTouch, Ifix, WinCC, RS View, Cimplicity, Citect	Advanced	3	Up to 12
4	**Distributed Control System (DCS)** Emerson, Siemens, Microverse, Honeywell, Yokogawa	Advanced	4	Up to 12
5	**AC Drive** Allen Bradley, Siemens, Schneider, GE Fanuc, Mitsubishi, Yaskawa, ABB	Advanced	2	Up to 12
6	**Servo Drives** Allen Bradley, Siemens	Advanced	2	Up to 12
Maximum 10 to 12 Participants for better understanding & good quality of training				

Industrial Communication & Networks Training



SN	Training Module	Level	Days	Participants In a Batch
7	**Industrial Communication & Networks Training**	Advanced	7	Up to 12
Maximum 10 to 12 Participants for better understanding & good quality of training				

On-site training



It is our opinion that customer training is considered one of the most important parts of any project. In line with our Total Quality Management and training policies, Prolific is able to provide a complete customer training program, which ensures full knowledge transfer.

Exact training requirements, however, can vary Considerably from customer to customer. For this reason, Prolific offers both standard training courses and customer specific training programs.

To help you decide which course is more suitable for you, we have assigned experienced technical support professionals to help you with your training requirements.

Programmable Logic Controllers (PLC)- Basic (Programming & Maintenance)

Duration of the course: 3 days

Hardware covered (any of the given)

Allen Bradley: Micrologix series, SLC 500	Siemens: Series S7 200 and S7 300
Schneider Electric: Modicon TSX Micro and Premium	GE Fanuc: Versa Max Series
Mitsubishi: Fx Series	Messung Systems: Nextgen

Topics Covered

Introduction to PLC

Topic	Mode
➢ Introduction to PLC hardware and role in automation ➢ Architectural Evolution of PLC ➢ Introduction to the field devices attached to PLC	Presentation and physical observations
➢ PLC Fundamentals - (Block diagram of PLC's) ➢ Detail information about PLC components o Power supply, CPU, I/Os, Communication bus ➢ Various ranges available in PLC's	Practical demonstration on hardware
➢ Types of Inputs & outputs / Source Sink Concepts ➢ Wiring of the I/O devices	Practical on PLC's
➢ Concept of flags ➢ Scan cycle execution	Practical on PLC's

Operation and Maintenance of PLC

Topic	Mode
➢ Setting up PLC's / Connecting CPU, I/O modules, Rack, Back plane and Communication bus	Practical – Assembling PLC
➢ Connecting Field devices to PLC's I/Os	Practical -Physical wiring
➢ Installing and to starting the programming terminals	Practical – Installation of software
➢ Identifying the status of PLC and communication bus	Practical on PLC's
➢ Fault detection and error handling	Practical on PLC's
➢ Forcing of the I/O's	Practical on Software
➢ Back up of the programs and reloading	Practical on PLC's
➢ CPU, I/O module replacements	Practical on PLC's

Programming of PLC

Topic	Mode
➢ Introduction to PLC programming software	Practical – Programming software
➢ Addressing concepts	Practical – Programming software
➢ Introduction to bit, byte & word concepts	Practical – Programming software
➢ Programming instructions arithmetic and logical ▪ Load /and /or/out / and Read / Write ▪ Compare / Add / Sub /And /Or – Blocks ▪ Leading edge / trailing edge instructions ▪ MOVE block application ▪ Timer and Counter Blocks programming ▪ Advanced instructions ▪ File handling ▪ Comment functions ➢ Master control /set /reset function	Practical – Programming software
➢ Upload, download, Monitoring of programs	Practical – Programming software
➢ Monitoring / Modifying data table values	Practical – Programming software
➢ Standard procedure to be followed in wiring / writing ladder etc	Practical – Programming software
➢ Troubleshooting and fault diagnostics of PLC	Practical – Programming software
➢ Documenting the project	Practical – Programming software

Programmable Logic Controllers (PLC)- Advance
Allen Bradley - ControlLogix Systems

Duration of the course - 4 days

Topics Covered

Topic	Mode
➢ Introduction to PLC hardware and role in automation ➢ Architectural Evolution of PLC ➢ Introduction to the field devices attached to PLC	Presentation and physical observations
➢ PLC Fundamentals - (Block diagram of PLC's) ➢ Detail information about PLC components o Power supply, CPU, I/Os, Communication bus ➢ Various ranges available in PLC's	Practical demonstration on hardware
➢ Identifying Logix System hardware and software Components	Practical demonstration on hardware and software
➢ Types of Inputs & outputs / Source Sink Concepts ➢ Wiring of the I/O devices	Practical on PLC's
➢ Concept of flags ➢ Scan cycle execution	Practical on PLC's

Operation and Maintenance of PLC

Topic	Mode
➢ Setting up PLC's / Connecting CPU, I/O modules, Rack, Back plane and Communication bus	Practical – Assembling PLC
➢ Connecting Field devices to PLC's I/Os ➢ Configuring Local 1756-I/O Modules	Practical -Physical wiring
➢ Connecting a Computer to a Communications Network	
➢ Installing and to starting the programming terminals ➢ Creating & Modifying an Rslogix new project ➢ Transferring a Project File to a Logix5000 Controller	Practical – Installation of software
➢ Identifying the status of PLC and communication bus	Practical on PLC's
➢ Creating Tags & Monitoring Data in an Rslogix 5000 Project ➢ Forcing of the I/O's	Practical on Software
➢ Managing Rslogix 5000 Project Files ➢ Back up of the programs and reloading	Practical on PLC's

Programming / Project Development

Topic	Mode
➢ Creating various Tasks ➢ Organizing Data ➢ Entering, Editing, & Verifying Ladder Logic ➢ Configuring Logix5000 Controllers to Share Data ➢ Configuring a Message ➢ Documenting & Searching for Project Components	Practical – Programming terminal
Ladder Logic Programming/Interpretation ➢ Identifying Programming Strategies & Techniques ▪ Documenting & Searching Ladder Logic ▪ Programming Basic Instructions ▪ Programming Timer & Counter Instructions ▪ Programming Program Control Instructions ▪ Programming Compare Instructions ▪ Programming Compute & Math Instructions ▪ Programming Move Instructions ▪ Programming with Advance Instruction Set	Practical – Programming terminal

➢ **Function Block Programming** ▪ Creating a Function Block Diagram ▪ Programming Logical Function Block Instructions ▪ Programming Timer & Counter Function Block Instructions ▪ Programming Analog Function Block Instructions ▪ Programming Timing Modes in a Function Block Instruction ▪ Programming & Monitoring an RMPS (Ramp/Soak) Function Block Instruction ▪ Controlling Program Flow Using Function Block Instructions	Practical – Programming terminal
➢ **Introduction to industrial networking** ➢ Identifying Industrial Networks for Use in a Logix5000 System	Theory session
➢ **Introduction to DeviceNet, EtherNet & DHRIO** ▪ Devicenet configuration & use. ▪ EtherNet configuration & use ▪ DHRIO Configuration & use ▪ Messaging over networks	Practical – Demonstration on network
ControlNet Configuration, Scheduling & Troubleshooting ▪ Basics of networks ▪ Modules & Media ▪ Addressing of nodes ▪ Scheduling ▪ Downloading/ uploading schedule ▪ Troubleshooting	Practical – Demonstration on network

Maintenance and Troubleshooting

Topic	Mode
ControlLogix Troubleshooting ▪ Interpreting Project Organization & Execution ▪ Monitoring Arrays & Tags of User-Defined Data Types ▪ Editing Ladder Logic Online ▪ Troubleshooting Controller Problems ▪ Monitoring GSV/SSV Instructions ▪ Forcing I/O & Toggling Bits ▪ Troubleshooting I/O Module Problems ▪ Troubleshooting Power Supply Problems ▪ CPU, I/O module replacements ▪ Fault detection and error handling	Practical on PLC

Programmable Logic Controllers (PLC)- Advance
Siemens – Step 7 300 and 400 Series PLC's

Duration of the course - 4 days

Topics Covered

Topic	Mode
➢ Introduction to PLC hardware and role in automation ➢ Architectural Evolution of PLC ➢ Introduction to the field devices attached to PLC	Presentation and physical observations
➢ PLC Fundamentals - (Block diagram of PLC's) ➢ Detail information about PLC components o Power supply, CPU, I/Os, Communication bus ➢ Various ranges available in Siemens S7 200 / 300 /400	Practical demonstration on hardware
➢ Types of Inputs & outputs / Source Sink Concepts ➢ Wiring of the I/O devices	Practical on PLC's
➢ Concept of flags ➢ Scan cycle execution	Practical on PLC's

Programming / Project Development

Topic	Mode
➢ SIMATIC S7 PLC Functionalities ➢ Setting up Siemens PLC hardware	Practical on PLC's
➢ Programming software ▪ PLC program structure in Siemens ▪ The instruction set of STEP 7 ▪ Parameters, functions and tools	Practical – PLC programming software
➢ CPU configuration, setting parameters and application of several I/O cards using the STEP 7 software	Practical – PLC programming software
➢ Working online / offline ➢ Starting S7 400 system ➢ Cold, Warm, and Hot Restarts ➢ Starting Up a PROFIBUS-DP Subnet ➢ Connecting remote I/Os using Profibus communication	Practical – PLC programming software
➢ Addressing Concepts in Siemens PLC's ➢ Detail information about Organizational Blocks, Function Block, Functions, System Function Block, System Function, Data block	Practical – PLC programming software
➢ Creating and Editing PLC programs ➢ Introduction to Bit Byte and Word Concept	Practical – PLC programming software
➢ Programming instructions arithmetic and logical ▪ Load /and /or/out / and Read / Write ▪ Compare / Add / Sub /And /Or – Blocks ▪ Leading edge / trailing edge instructions ▪ MOVE block application	Practical – PLC programming software
➢ Programming instructions arithmetic and logical ▪ Timer Blocks programming ▪ Counter Block programming ▪ Comment functions ▪ Comments in the PLC programs ▪ Handling Analog I/Ps ➢ Conversion shift and jump instructions	Practical – PLC programming software

Advanced Programmers ▪ Analyse /evaluate of status words ▪ Jump instructions ▪ Accumulator functions ▪ Extended mathematical functions ▪ Indirect Addressing ▪ Complex Data-Types ▪ Extended S7 Basic communication ▪ PID / Control functions	Practical – PLC programming software
Siemens Networking ▪ Industrial Data Networks ▪ Industrial Ethernet – General ▪ Siemens Profibus ▪ Communication via a Multi Point Interface (MPI)	Practical – on PLC's

Programming / Project Development

Troubleshooting and Maintenance procedure ▪ Replacing Digital or Analog Modules ▪ Replacing CPUs ▪ Replacing the Backup Battery ▪ Replacing a Power Supply Module ▪ Back up and Restoring programs ▪ Fault diagnostics and error handling	Practical – on PLC's

Programmable Logic Controllers (PLC)- Advance
Schneider Quantum PLC's

Duration of the course - 4 days

Topics Covered

Topic	Mode
➢ Introduction to PLC hardware and role in automation ➢ Architectural Evolution of PLC ➢ Introduction to the field devices attached to PLC	Presentation and physical observations
➢ PLC Fundamentals - (Block diagram of PLC's) ➢ Detail information about PLC components o Power supply, CPU, I/Os, Communication bus	Practical demonstration on hardware
➢ Types of Inputs & outputs / Source Sink Concepts ➢ Wiring of the I/O devices to the PLC's	Practical on PLC's
➢ Concept of flags ➢ Scan cycle execution	Practical on PLC's

Schenider PLCs

Topic	Mode
➢ Introduction to various ranges available in Schneider PLCs	Presentation
➢ Setting up Schneider hardware system	Practical on PLC's
➢ Configure and layout a Modicon system, including input/output modules and local and remote input/output systems ➢ Addressing concepts in Modicon	Practical – PLC programming software
➢ Introduction to PL 7 PLC programming software ➢ Configuration, Program, Variables ➢ Hardware and Software Configuration	Practical – PLC programming software
➢ Ladder Programming language and methodology ➢ Creating a new ladder logic program ➢ Handing Digital instructions (NO , NC, Rising Edge, Falling Edge, Coils, Set, Reset)	Practical – PLC programming software
➢ Working with Timer and Counters ➢ Register, Drum ➢ Horizontal, Vertical and Operate Block	Practical – PLC programming software
➢ Scaling and PID instructions ➢ Creating subroutine ➢ Jump instruction	Practical – PLC programming software
➢ Introduction to Grafset ➢ Creating an application using Grafset	Practical – PLC programming software
➢ Task and event management ➢ Defined function blocks	Practical – PLC programming software
➢ Safety considerations related to your PLC system	Practical – PLC programming software
➢ Networking ▪ Industrial Data Networks supported by Schneider ▪ Industrial Ethernet – General ▪ Modicon Modbus and Modbus + networking protocols	Practical on PLC's

Troubleshooting

Topic	Mode
➢ PLC trouble shooting techniques ▪ Distinguish between hardware problems, software problems, and field problems ➢ Application debugging ➢ Replacing the faulty modules ➢ Fault diagnostics and error handling	Practical on PLC's

Distributed Control Systems (DCS)- Advance
Emerson – Delta V / Yokogawa CS 3000 / ABB Freelance AC800F

Duration of the course: 4 days

Topics Covered

DCS General

Topic	Mode
➢ Introduction to various control systems used ➢ Applications of DCS system	Presentation
➢ Architectural evolution for DCS ➢ Hardware architecture of DCS ▪ Processor ▪ Power supply ▪ I/O modules ▪ Communication bus ▪ Operator and Engineering station ▪ Redundancy	Presentation
➢ Comparison between the hardware PLC & DCS	Presentation

Practical exposure on DCS System

➢ Practical exposure on DCS system ➢ Overview of the system components and topologies ➢ Detailed information on ▪ Controllers ▪ I/O cards ▪ Carriers ▪ System power suppliers ▪ Control networks	Practical on DCS
➢ Designing DCS based automation solution for the application	Practical on DCS
➢ Setting up DCS system with PS, Controllers, I/Os with communication bus system ➢ Connecting field devises to DCS	Practical on DCS
➢ Configuring engineering and operator stations	Practical on DCS
➢ Application development in DCS system ▪ Graphical User Interface Development ▪ Control Modules ▪ Workspace	Practical on DCS
➢ Alarm and Process History ➢ Sequential Function Charts ➢ Display Scripting	Practical on DCS
➢ Users and Securities ➢ Custom Faceplates ➢ Custom Dynamos	Practical on DCS
➢ Scripting and programming languages	Practical on DCS
➢ Configuring nodes on networks	Practical on DCS
➢ Networking technologies in DCS systems	Practical on DCS
➢ Redundancy systems available in DCS ▪ CPU redundancy ▪ Power supply redundancy ▪ Communication redundancy ▪ I/O redundancy	Practical on DCS

Troubleshooting

Topic	Mode
➢ DCS trouble shooting techniques ▪ Distinguish between hardware problems, software problems, and field problems ➢ Reading the Error status bits ➢ Replacing the faulty modules ▪ CPU, I/O Modules / Power supply ➢ Fault diagnostics and error handling ➢ Backup & Restore Procedures and Techniques	Practical on DCS

Distributed Control Systems (DCS)- Advance
Siemens – PCS 7

Duration of the course: 4 days

Topics Covered

DCS General

Topic	Mode
➢ Introduction to various control systems used ➢ Applications of DCS system	Presentation
➢ Architectural evolution for DCS ➢ Hardware architecture of DCS ▪ Processor ▪ Power supply ▪ I/O modules ▪ Communication bus ▪ Operator and Engineering station ▪ Redundancy	Presentation
➢ Comparison between the hardware PLC & DCS	Presentation and physical observations

Practical exposure on Siemens PCS 7 System

➢ Practical exposure on Siemens PCS 7 system ➢ Overview of the system components and topologies ➢ Detailed information on ▪ Controllers ▪ I/O cards ▪ Carriers ▪ System power suppliers ▪ Control networks	Practical on DCS
➢ Designing PCS 7 DCS based automation solution for the application	Practical on DCS
➢ Setting up PCS 7 system DCS system with PS, Controllers, I/Os with communication bus system ➢ Connecting field devises to PCS 7 System	Practical on DCS
➢ Configuring engineering and operator stations	Practical on DCS
➢ Operator Station – OS (WinCC HMI) ▪ WinCC Global scripts. Diagnostics, triggering and examples. ▪ Alarming ▪ Trending	Practical on DCS
➢ Application development in DCS system ▪ Graphical User Interface Development ▪ Control Modules ▪ Workspace	Practical on DCS
➢ Alarm and Process History ➢ Sequential Function Charts ➢ Display Scripting	Practical on DCS
➢ Users and Securities ➢ Custom Faceplates ➢ Custom Dynamos	Practical on DCS
➢ Scripting and programming languages ▪ Continuous Control – CFC ▪ Structured Control Language – SCL ▪ Sequential Control – SFC	Practical on DCS
➢ Configuring nodes on networks	Practical on DCS
➢ Networking technologies in PCS 7 systems ▪ Introduction to - Industrial Ethernet ▪ Introduction to Profibus technology	Practical on DCS
➢ Redundancy systems available in DCS ▪ CPU redundancy ▪ Power supply redundancy ▪ Communication redundancy ▪ I/O redundancy ➢ Management of Redundancy ▪ External or Remote SP control	Practical on DCS

Troubleshooting

Topic	Mode
➢ DCS trouble shooting techniques ▪ Distinguish between hardware problems, software problems, and field problems ➢ Reading the Error status bits ➢ Replacing the faulty modules ▪ CPU, I/O Modules / Power supply ➢ Fault diagnostics and error handling ➢ Backup & Restore Procedures and Techniques	Practical on DCS

Supervisory Control and Data Acquisition Software (SCADA) - Advance WonderWare InTouch / Siemens Win CC / Allen Bradley RS View / GE Fanuc Cimplicity / Citect

Duration of the course: 3 days

Objectives
Participants will learn selection, programming, installation, commissioning and troubleshooting of SCADA software

SCADA General

Topic	Mode
➢ Applications of SCADA software ➢ Different packages available with I/O structure ➢ Features of SCADA software	Practical

Practical exposure on SCADA software

Topic	Mode
➢ Creating a new SCADA application ➢ Creating Database of Tags	Practical – Software
➢ Creating & Editing graphic display with animation ▪ Data Entry / Start Stop command ▪ Analog entry ▪ Sizing, Movement, Blinking, Visibility, Filling	Practical – Software
➢ Trending ▪ Creating & Accessing Real-time and Historical Trends	Practical – Software
➢ Creating Alarms & Events	Practical – Software
➢ Writing logic through script	Practical – Software
➢ Connectivity with the different hardware ▪ Communication protocols ▪ Communication with PLC ▪ Communication with Data Acquisition System	Practical – Software
➢ Connectivity between software	Practical – Software
➢ Concept of DDE, DLL, OPC drivers	Practical – Software
➢ Commissioning the network nodes	Practical – Software
➢ Troubleshooting the application ▪ Fault diagnostics and error handling ▪ Sorting communication problems	Practical – Software

Drives - Advance
Allen Bradley / Schneider / Yaskawa / Siemens / ABB / Delta

Duration of the course: 2 days

Objective
Participants will be able to select, program, install and maintain drives to suit their application.

Topic	Mode
➢ Introduction to AC Drives	Practical – Drives setup
➢ Selection criteria of the drives for particular application ▪ Speed /Torque Characteristic ▪ Frequency converter techniques. ▪ Control techniques of AC drives	Presentation
➢ Configuration of different motor parameter for drives ▪ Speed and torque control ▪ Types of braking ▪ Ramp selection ▪ Jog/inch selection ▪ Flux optimization ▪ No. Of poles ▪ Load equalization ▪ Selection of slip & skip frequencies ▪ Preset speed selection ▪ RPM etc.	Practical – Drives setup
➢ Remote and Local operation of Variable Speed Drives	Practical – Drives setup
➢ Communication with PLCs	Practical – Drives setup
➢ Study of different operational methods ▪ PWM method ▪ Flux vector control ▪ Vector voltage control ➢ Direct torque control etc. on which Variable Speed Drives work	Practical – Drives setup
➢ Troubleshooting	Practical – Drives setup
➢ Case study and different applications of Drives in the industry	Practical – Drives setup



Industrial Communication & Networks Training

Duration of the course: 7 days

Remarks

If a complete group of participants is interested in a specific Communication & Networks application, the course contents can be adapted accordingly with agreement of the customer. If customers are normally working with specific Communication & Networks products this will be taken into account while practicing through exercises.

Topics Covered

- Industrial Data Networks: **1 Day**
- Industrial Ethernet – General: **1 Day**
- Siemens Profibus: **1 Day**
- Siemens Industrial Ethernet: **1 Day**
- Rockwell DeviceNet/ControlNet/Ethernet IP: **2 Days**
- Modicon Modbus: **1 Day**

➢ Detailed Course Contents

Industrial Data Networks

- Introduction to Industrial networking
- Basic network knowledge, supported by demonstrations and practical assignments;
- Data communication and networking used in plants
- Serial connections: RS232/RS422/RS485
- Sensor and Control-level Networks
- Introduction to network equipment, Routing, Switching
- Introduction to commonly used networking protocols

Industrial Ethernet – General

- Overview of common industrial communication networks.
  - OSI model
  - Available networks (Master/Slave, Client/Server and producer/Consumer)
  - Communication relations (Point to Point, Multicast and Broadcast)
  - Transmission media (Cables, Wireless)
  - Network Standards

- Common network topologies and media
  - The history of Ethernet
  - Standards (IEEE 802.3 and 802.11)
  - Transmission media and 10BASE-5, 10BASE-2, 1BASE-T, 10BASE-T, 10BASE-F, Fast Ethernet, Gigabit Ethernet and 10-Gigabit Ethernet
  - Ethernet and the OSI model
  - Cables, connectors and Ethernet devices: Repeaters, Hubs, Bridges, Routers, Switches, Gateways, etc.
  - Industrial Ethernet and OPC servers
  - Industrial Ethernet as a fieldbus and its requirements
  - IP, TCP and UDP ground rules
  - Industrial Ethernet protocols; Modbus/TCP, ProfiNet, Ethernet/IP, Foundation Fieldbus HSE, IDA
  - Organizations (IAONA, ODVA and IDA)

Siemens Profibus

- Profibus as open fieldbus system – relationship to the ISO/OSI model
- Transmission medium: RS485, fibre-glass and/or IEC 1158-2 technology
- Physical characteristics and installation details
- Profibus services: FDL, DP, FMS and PA
- Data link functions: SDA, SRD, SDN and CSRD
- Token passing principle: TTH, TTR, TRR parameters…
- Addressing: Profibus address (FDL) and Service Access Point (SAP)
- Active versus passive Profibus partners
- FMS service details:
    - Client server model
    - Virtual Field Device (VFD)
    - Object dictionary (OD)
    - Communication Relations List (CRL)
    - Standard and user defined data types

- DP service details:
    - Master slave model
    - Remote I/O system: distributed peripheral
    - Multi-DP sub-network on one cable

Siemens Industrial Ethernet

Contents

- Overview of common industrial communication networks.
- Common network topologies and -media.
- Access methods to the network media: token passing, collision detection etc.
- Master-Slave, Client-Server and Peer-to-Peer concepts.
- The 7 layer OSI model from the ISO.
- Historical aspects of the Siemens Industrial Ethernet.
- The OSI-stack-profile of the Siemens Industrial Ethernet.
- Transport layer ISO-8073 protocol.
- Comparison of ISO-8073 and TCP/IP.
- The Sinec-AP protocol.
- Application of “handling blocks” in the S7program.
- Practical demonstration on connecting remote I/Os on ethernet

Rockwell DeviceNet / ControlNet /Ethernet IP

Specific contents

- General overview of the various Rockwell networks

Part 1: ControlNet

- Familiarizing with ControlNet protocols
- Installing ControlNet networks (hardware)
- ControlNet network configuration with RSNetworx for ControlNet
- Planned and unplanned data-exchange with ControlNet
- Downloading and uploading projects
- Network diagnostics / trouble shooting
- Practical demonstration of connecting devices on control net networks

Part 2: DeviceNet

- Familiarizing with DeviceNet protocols
- Installing DeviceNet networks (hardware)
- DeviceNet network configuration with RSNetworx for DeviceNet
- Downloading and uploading projects
- Explicit-messaging
- Network diagnostics / trouble shooting
- Practical demonstration of connecting devices on Device net networks

Part 3: Ethernet IP

- Familiarising with Ethernet TCP/IP and EtherNet IP
- Installing EtherNet IP networks (hardware)
- Data-exchange over EtherNet IP
- Data-exchange over measurement and control devices
- Network diagnostics / trouble shooting
- Practical demonstration of connecting devices on Ethernet IP networks

Modbus

Contain

- INTRODUCTION
  - Overview of Modbus and development
  - How Modbus fits into the RS-232/ RS-485/Ethernet and TCP/IP Standards
  - List of typical problems
- OVERALL METHODOLOGY
  - Common symptoms, problems and solutions
  - How to quickly identify likely causes
  - Overall basic steps
  - Communications issues
  - Grounding/shielding/noise
- BASIC STANDARDS
  - **RS-232**
    - Fundamentals
    - Problems: cabling, male/female, DTE/DCE, 9-pin, handshaking, voltages, noise, isolation
    - RS-232 Practical Troubleshooting Session
  - **RS-485**
    - Fundamentals
    - Problems: cabling, common mode voltage, converters, isolation, idle state, terminations, control - Hardware/software
    - RS-485 Practical Troubleshooting Session
- MODBUS OVER ETHERNET
  - Modbus over TCP/IP
  - Modbus-IDA Packet structure
  - Typical troubleshooting issues
  - Practical session with Modbus over TCP/IP
- INDUSTRIAL AUTOMATION STANDARDS:
  - **Modbus over Serial Communications (such as RS-232/RS-485)**
    - Fundamentals
    - Problems: no response, exception reports, noise, radio interfaces, physical and application layers
    - Modbus troubleshooting session
  - **Modbus Plus**
    - Fundamentals
    - Problems: cabling, grounding, shielding, terminators, token passing